Area Between Curves

Section 1: Vertical Slices (Integration with Respect to x)

1 Find the area enclosed by $y = x^2$ and $y = 2x - x^2$.

a. 1/3

b. 2/3

c. 1/6

d. 1

2 Find the area between $y = x^2$ and $y = 4$.

a. 32/3

b. 16/3

c. 8

d. 64/3

3 Find the area between $y = \sqrt{x}$ and $y = x^2$.

a. 1/3

b. 2/3

c. 1/6

d. 1/2

Section 2: Horizontal Slices (Integration with Respect to y)

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4 Find the area enclosed by $x = y^2$ and $x = y + 2$.

a. 9/2

b. 16/3

c. 7/2

d. 9

5 Why is integrating with respect to y advantageous for curves like $x = y^2 - 4$ and $x = 0$?

a. The right curve minus left curve is a single continuous expression without piecewise splitting

b. Integration with respect to x is impossible

c. It changes the area value

d. The bounds are always 0 and 1

6 Find the area enclosed between $x = 4 - y^2$ and the y-axis.

a. 32/3

b. 16/3

c. 8

d. 16

Section 3: Crossing Curves and Multiple Regions

7 Find the total area enclosed between $y = \sin x$ and $y = \cos x$ from $x = 0$ to $x = \pi/2$.

a. $2\sqrt{2} - 2$

b. $\sqrt{2} - 1$

c. 2

d. 1

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8 Find the area enclosed between $y = x^3 - x$ and $y = 0$ for $x \in [-1, 1]$.

a. $1/2$

b. 0

c. $1/4$

d. 1

9 Find the area enclosed between $y = x^3$ and $y = 4x$.

a. 8

b. 4

c. 16

d. 0

Section 4: Applications and Boundaries

10 Find the area of the region bounded by $y = e^x$, $y = e^{-x}$, and $x = 1$.

a. $e + 1/e - 2$

b. $e - 1/e$

c. $e + 1/e$

d. $e - 2$

11 Find the area bounded by $y = \frac{1}{x}$, the x-axis, $x = 1$, and $x = e^2$.

a. 2

b. 1

c. e^2

d. $2e$

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12 The Gini index measures income inequality as twice the area between the line of perfect equality $y = x$ and the Lorenz curve $L(x) = x^2$ on $[0, 1]$. Find the Gini index.

a. $1/3$

b. $1/6$

c. $2/3$

d. $1/2$

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